# 40-year annual time series of dominant tree species and leaf type cover for Italy derived from Landsat using deep learning

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**Abstract.** Consistent long-term maps of forest functional type and dominant tree species cover at high spatial resolution are currently unavailable for Italy. The study presents annual 30 m resolution maps of continous evergeen broad-leaved, deciduous, and coniferous leaf type cover fractions and dominant tree species class probabilities for all forested pixels in Italy spanning 1987–2025, derived from the Landsat Collection-2 Level-2 archive (Landsat 4/5/7/8/9). Two Time Series Transformer models are trained via a three-stage workflow combining contrastive pretraining with supervised fine-tuning on Italian forest plot observations, forest inventories, and a synthetically generated training dataset to predict both tasks. Pixel-level uncertainty estimates are provided as the standard deviation of a five-seed deep ensemble for both models. A three-year observation pooling strategy is applied to increase cloud-free observation density and mitigate inter-annual variability. Evaluation of the time series maps are based on hold-out areas and a comparison with historical orthophotos. The 39-year time series enables historical reconstruction of forest composition dynamics and long-term ecological change detection at national scale and relevant time scales for climate-driven species range shifts.



## 1 Introduction

**Landsat for long term forest monitoring:** With nearly 40 years of continuous acquisitions, the Landsat archive constitutes the longest consistent Earth observation record available for global vegetation monitoring (Wulder et al., 2022). Its four decades of continuity makes it uniquely suited for change detection or hindcasting of environmental variables including forest habitat characteristics (Bell et al., 2024; Turubanova et al., 2023). The archive has been applied to discriminate forest types and quantify forest variables such as canopy height or greening trends from regional to continental scales (Pflugmacher et al., 2019; Obuchowicz et al., 2024; Yel et al., 2026). In monitoring long-term changes and disturbances in forest vegetation under climate change, the Landsat afchive provides a valuable modelling base (Viana-Soto and Senf, 2025; Grünig et al., 2026). The relatively slow change and succession dynamics of forest vegetation, which is unfolding over decades, make this long temporal availability particularly valuable. A shorter observation period would fail to capture natural stand-level transitions such as species shifts or forest densification, which happen gradually over long time windows (Tong et al., 2023; Midolo et al., 2026). The medium spectral, spatial, and temporal resolution of Landsat (6 reflectance bands, 30 m ground sampling distance, and 8–16 day revisit time) is adequate for capturing annual phenological dynamics at stand level, which demonstrated to be critical for distinguishing tree species and functional leaf types (Kollert et al., 2021; Hemmerling et al., 2021; Wang et al., 2022).

**Challenges:** Despite these strengths, the use of multidecadal Landsat data for vegetation mapping raises several methodological challenges. The number of usable cloud-free observations has increased since the early periods of the archive, due to the addition of new Landsat satellites. This systematic increase in observation density has been shown to cause spurious trends in annual maximum NDVI, where up to 50% of observed Landsat-based greening in alpine environments can be attributed to observational sampling bias rather than true vegetation change (Bayle et al., 2024). A second challenge arises from the failure of the Scan Line Corrector (SLC-off) on Landsat 7 ETM+, which introduces systematic spatial data gaps. Finally, spectral radiometric differences between sensors — arising from different sensors, band widths, and data processing procedures across Landsat missions — introduce inter-sensor discontinuities that may produce spurious changes in modelled variables across the 40-year record.

**ML/DL for dominant tree species and leaf type cover mapping:** Multitemporal satellite data consistently outperform single-date or mosaic-based approaches for tree species and leaf type mapping. Phenologically informative seasons — particularly leaf-off periods and autumn senescence — contribute disproportionately to species discrimination (Kollert et al., 2021; Tan et al., 2025). When sufficient temporal coverage is available, dense time series exploiting the full phenological cycle can outperform the inclusion of additional environmental predictor variables such as topography or climate indices (Hemmerling et al., 2021). These findings highlight temporal resolution and coverage as primary sources for improving species-level modelling and mapping performances.

Deep learning, and in particular Transformer-based architectures, provide a framework for processing raw, irregular, multi-band satellite time series (STS). Transformers process sequences of observations without requiring interpolation to a regular sequence, an essential property for long Landsat archives with varying observation density across decades and sensor platforms (Zerveas et al., 2020; Yuan et al., 2022). Self-supervised pretraining on large unlabelled STS or image time series datasets has been shown to improve representation quality and performance on downstream tasks, particularly in data-scarce settings (Yuan Yuan et al., 2020; Hiebl et al., 2025; Feng et al., 2026). Contrastive learning approaches exploit temporal and seasonal variability by training representations to be invariant across different observation windows of the same location — a property suited to stable land cover types such as forest stands (Mañas et al., 2021; Chen et al., 2020). Providing calibrated uncertainty estimates alongside predictions has emerged as an important requirement for ecological mapping applications, to enhance interpretability by users. Deep ensembles of independently trained networks deliver calibrated epistemic uncertainty that reliably identifies out-of-distribution inputs and unreliable predictions, when feature and label related domain shifts appear during the mapping phase (Sylvain et al., 2024; Hiebl et al., 2025).

**Relevance for Italys forests:** Italian forests are undergoing rapid and spatially complex changes. The Italian National Forest Inventory (INFC2015) documents a continuing expansion of forested area from approximately 10.46 million ha in INFC2005 to 10.98 million ha in INFC2015 driven by land abandonment and targeted reforestation programmes (Gasparini et al., 2022). At the same time, land abandonment in combination with climate change alters phenological characteristics and forest composition in favor of species with higher drought resilience (Herraiz et al., 2025; Chelli et al., 2017), while evergreen broad-leaved species are spreading in deciduous forest areas (Conedera et al., 2018). These processes are spatially heterogeneous, reflecting regional variety in land use changes, climate change, and historical species composition. Despite these dynamics, spatially explicit maps of forest composition and variables in Italy remain restricted to static maps with no continuous temporal resolution. The most recent national reference product, the Carta Forestale d'Italia 2020 (CFI2020), provides a single-epoch wall-to-wall forest map at 1:10,000 scale based on photo-interpretation of aerial orthophotos, but does not capture interannual dynamics or historical trajectories of change (Mattioli et al., 2025).

**What will be done?** Our study will fill this gap by providing data-driven, spatio-temporal continuous maps of leaf type cover (LTC) fractions and dominant tree species (DTS) probabilities for all forested areas in Italy - annually and at Landsat-native $30m$ resolution. We assemble and process 40-years of Landsat Collection-2 Level-2 time series from 1985 to 2025 (Landsat-4/5/7/8/9) and train multiple Time Series Transformer models (TSTpad) via contrastive pretraining and a fine-tuning workflow. The final product are annual $30m$ maps of i) 11 dominant tree species classes, and ii) LTC fractions for evergreen broad-leaved (EVE), deciduous broad-leaved (DEC) and coniferous (CON) functional leaf type. Additional ensemble-based uncertainty estimations will be provided per-pixel and target variable.

## 2 Materials and Methods

### 2.1 Study Area

The study covers all forested areas of peninsular and insular Italy, spanning several biogeographical, climatic, and latitudinal regions and the full elevation gradient of the country (Fig. 1). Forest extends from sea level — $34.1\,\%$ of the national forest area lies below $500\,\mathrm{m}$ a.s.l. and $1.1\,\%$ above $2000\,\mathrm{m}$ a.s.l. (Gasparini et al., 2022) — up to the alpine treeline, which in the western Italian Alps was dated at

2505 m a.s.l. in 2000 and 2515 m a.s.l. in 2008 (Leonelli et al., 2011). Italian forests fall within alpine, temperate, sub-Mediterranean, and Mediterranean climate zones (Chelli et al., 2017), and are dominated by four forest habitat types: broadleaved-deciduous and temperate coniferous forests concentrated in the Alps and Apennines, and evergreen broad-leaved and Mediterranean coniferous forests concentrated along the coasts and in the south (Bricca et al., 2026). This climatic and ecological heterogeneity is a central challenge for wall-to-wall mapping, as models trained predominantly on well-sampled regions risk out-of-domain predictions in climatically and ecologically different areas with sparser reference data. Forests cover 37% of Italy's territory (10.98 million ha) (Gasparini et al., 2022). Italy's position as a transition zone between the Mediterranean and the temperate climate of Central Europe makes its forests especially vulnerable to climate change: warming and increasing drought frequency are already documented to shift phenology and favour more drought-resilient and evergreen broad-leaved species at the cost of deciduous ones (Chelli et al., 2017; Herraiz et al., 2025; Conedera et al., 2018). These climate-driven shifts compound ongoing changes in forest management and land use, jointly reshaping tree species distribution and biodiversity across the country (Bricca et al., 2026).

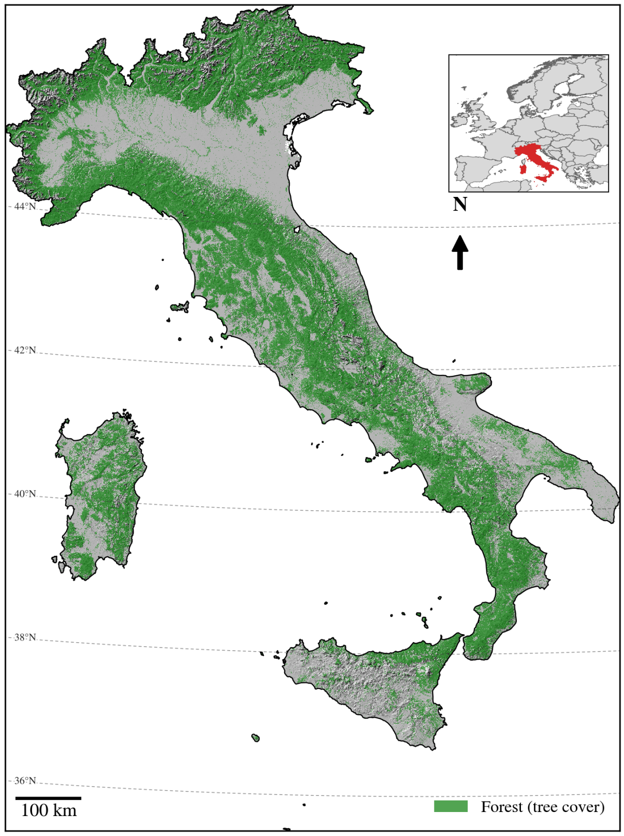


**Figure 1.** Study Area Italy with high forest cover in the Alps and mountainous areas of Central Italy.

## 2.2 Modelling Data

An overview of the used datasets during training are provided in Tab.1 and Fig.2.

### 2.2.1 Italian forest vegetation plot databases

The Forest Vegetation Database Italy (VDBI) is a compilation of over 16,000 phytosociological and vegetation relevé plot observations collected across Italy from the 1970s to the present. Each plot record contains species composition data, structural attributes, and cover estimates for the tree strata (Alessi et al., 2019). The VDBI covers the full range of Italian forest types from Mediterranean macchia to subalpine conifer belts. It provides temporal depth extending back to approximately 1972, enabling historical training data for pre-2000 Landsat time steps. The location accuracy of single plot observations in the dataset varies across observation year and source dataset. To avoid non-forest pixels leak into the training and test datasets, VDBI was cleaned manually based on orthophoto imagery. The final dataset after the cleaning process provides 9837 plot observations. Plot records include LTC estimates (EVE, DEC, CON) and tree species cover fractions, which were used to retrieve dominant DTS classification labels, allowing joint training of the regression ($VDBI_{DTS}$) and classification ($VDBI_{DTS}$) TST-pad model. A $10 \times 10$ km regular polygon grid was created for the train/validation/test split to reduce spatial autocorrelation between training and validation sets. A 20% validation split, stratified by polygon identifier is applied throughout all training stages (Safonova et al., 2023).

### 2.2.2 Field plot observations

The Vegetation Plot Observations (VPO) dataset consists of recent field campaigns conducted within 7 Italian National Parks or protected areas (see: Hiebl et al. (2025, 2026)). Each site comprises four systematic 100 m$^2$ vegetation plots arranged on a regular grid, within which species-wise cover is estimated for three vertical layers (0–1 m, 1–10 m, and above 10 m). LTC is extracted as sum of EVE, DEC, and CON species cover fractions, while DTS is - similar to VDBI - retrieved from species cover by selecting the tree species with highest cover fraction. The VPO data used in this study consists of observations from 2023 to 2025 field season, yielding a total of 2016 individual plot observations from 504 sites. The VPO dataset provides high-quality, spatially clustered reference data for the most recent Landsat time steps (post-2020) and serves as the primary validation reference for model performance in the recent high-density observation period. The VPO dataset is further split into a training set (80%) and a validation set (20%) to evaluate model performance. The split was performed spatially after clustering

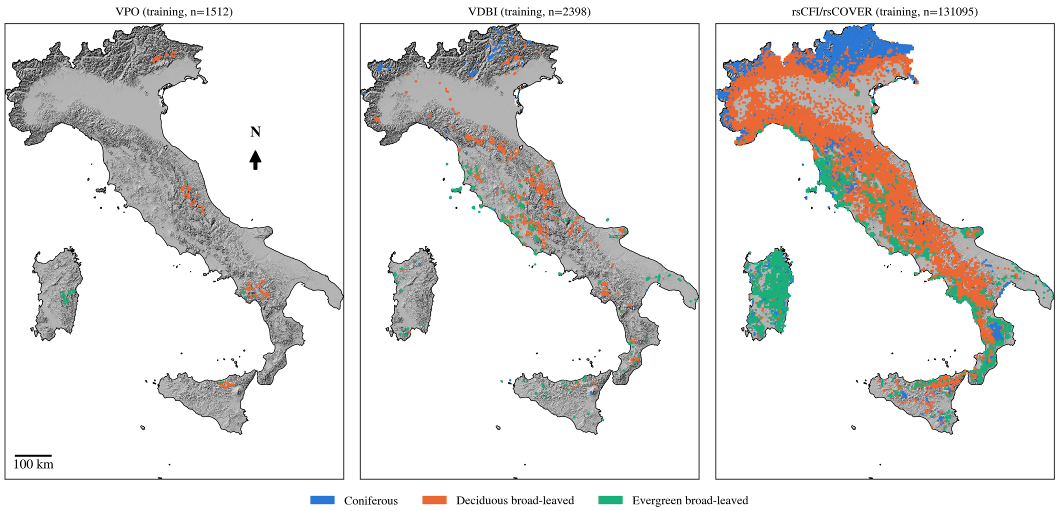


**Figure 2.** Overview of training datasets used in the study.

**Table 1.** Overview of datasets used for model training and validation.

| | n samples | LTC | DTS | |
|---|---|---|---|---|
| VPO | 2016 | yes | yes | conducted in the field 2023-2025, standardized $10 \times 10$ m plots (see Sec. 2.2.2) |
| VDBI | 2998 | yes | yes | observations with varying size, acquisition time, and location accuracy ($0.01 - 10$ km) (see Sec. 2.2.1) |
| rsCFI | 131095 | - | yes | randomly sampled across CFI dominant tree species (see Sec. 2.2.3) |
| csCFI | 30380 | - | yes | pure stand polygons sampled from CFI (see Sec. 2.2.3) |
| rsCOVER | 131095 | yes | - | same locations as rsCFI, synthetically created using AlphaEarth embeddings (see Sec. 2.2.4) |
| csCOVER | 30380 | yes | - | same locations as csCFI, synthetically created using AlphaEarth embeddings (see Sec. 2.2.4) |

the sites by their coordinates into 10 clusters per region. The spatial clustering was done to stabilize training and prevent over-optimistic accuracy estimates due to spatial autocorrelation (Safonova et al., 2023; Hiebl et al., 2025).

### 2.2.3 CFI Forest Inventory Data

The CFI2020 is the first national forest map of Italy produced at operational scale (1:10,000), based on photointerpretation of aerial orthophotos from 2018 to 2020 against three simultaneously applied forest definitions (Mattioli et al., 2025). The CFI2020 delineates forested areas across all 21 Italian regions and provides stand-level attributes including forest category, silvicultural system, and DTS group. The initial 43 unique classes in the dataset were collapsed into 11 dominant tree species classes: silver fir, spruce, pine, deciduous oak, chestnut, beech, manna ash - hornbeam, evergreen oak, coniferous others, deciduous others, evergreen others. In this study, the CFI2020 polygon layer serves two roles: i) as an additional training/validation data source for the DTS classification task, and ii) as a source of decision rules for cleaning the synthetic LTC training dataset (see 2.2.4). In the first case two different datasets are created by i) randomly sampling a set of evenly distributed points over all classes (rsCFI), and ii) by sampling points from manually delineated pure stand polygons that are evenly distributed across classes (csCFI). In the second case, CFI2020 species attribution (DTS) is used to impose ecologically plausible constraints on predicted cover fractions of the synthetic dataset (see 2.2.4).

### 2.2.4 Synthetic leaf type cover data

Spatially explicit wall-to-wall leaf type cover data for model training are not available for Italy. Following the approach of Kang et al. (2021) and Tan et al. (2025) — who demonstrate the use of coarser or indirectly derived data sources as proxy training targets where direct measurements are absent — an artificial leaf type cover dataset was created from the VDBI and VPO plot observations. A Random Forest regression model was trained on the combined VDBI and VPO training split, with EVE, DEC, and CON cover fractions as prediction targets. As input features, AlphaEarth embeddings (Brown et al., 2025; Hiebl et al., 2026) — multimodal spectral-spatial-temporal feature representations derived from a globally pretrained geospatial foundation model — were extracted for all training plot locations. AlphaEarth embeddings were shown to achieve high species classification accuracy with very limited training data in Italian mountain forests (Ball et al., 2026), making them particularly suited as inputs for the pseudo-label generation step.

The trained Random Forest model was then applied to the full CFI2020 forest extent to generate a nation-wide pixel-level pseudo-label layer for each cover fraction. The labels were extracted at the same locations as rsCFI (rsCOVER)and csCFI (csCOVER). This pseudo-label layer constitutes the primary continuous-cover target during the contrastive pre-training and early fine-tuning stages. To reduce label noise introduced by model prediction errors, the CFI2020 species attribution was used to apply ecologically motivated cleaning rules: e.g. pixels within pine-dominated forest polygons were required to have at least 60% CON cover and fewer than 30% EVE cover; analogous rules were applied to classes of beech, oak, and dominant tree species. These cleaning rules remove the most pseudo-label errors while preserving the spatial completeness of the training dataset. If a sampled synthetic plot does not meet the criteria the sample is dropped. This way we prevent ecologically non-feasible labels being forwarded into the final training loop.

### 2.2.5 Landsat time series

figure: landsat time series plot over all years for 2 different dominant tree species, include modelling approach with 3 year window aggregation

Annual Landsat time series were assembled for the full Italian national territory from Microsoft Planetary Computers STAC Catalog using Landsat Collection-2 Level-2 surface reflectance products from Landsat 4 TM (1982–1993), Landsat 5 TM (1984–2013), Landsat 7 ETM+ (1999–2023), Landsat 8 OLI (2013–present), and Landsat 9 OLI-2 (2021–present). Scenes with cloud cover higher than 75% were excluded, which led to a per year count of approx. 80 to 120 scenes. Per-pixel cloud, cloud shadow, cirrus, and snow masking was performed using the Landsat Collection-2 Level-2 QA PIXEL quality bitfield. Per-observation quality flags derived from QA PIXEL are propagated through the entire modelling pipeline as a key-padding mask, ensuring that cloud-contaminated or padded time steps are ignored by all attention layers of the TSTpad model. For each forested pixel, 14 spectral features are computed at each valid observation: six surface reflectance bands (blue, green, red, near-infrared, SWIR1, SWIR2) and eight derived spectral indices (NDVI, NIRv, NDMI, EVI, WDRVI, NBR, NDWI, GNDVI). At the training plot location the resulting multi-annual time series are stored as Zarr-compressed arrays to enable efficient random-access loading during model training. For mapping, the Landsat scenes were fetched and preprocessed on-the-fly in $25 \times 25$ km chunks from Microsoft Planetary Computer parallel to the GPU-based model prediction.

### 2.2.6 Additional data sources

The ESA WorldCover product is used as an initial broad-scale delineation of forested areas across Italy (Zanaga et al., 2022). Only pixels labelled as tree cover are forwarded for prediction in the final maps. We are aware that the ESA WorldCover product does not provide a sufficient forest mask for detailed analysis of the datasets, as many mixed pixels and edge cases are remaining. But by applying a conservative masking during mapping we leave room for stricter forest delineation through other products without loosing regional accuracy.

## 2.3 Time Series Transformer Architecture

The backbone model is an encoder-only Time Series Transformer extended to support irregular, sparse, annual satellite image time series (TSTpad). The architecture employs multi-head self-attention across the temporal dimension, enabling the model to learn both short-range phenological features and long-range inter-seasonal dependencies within the aggregated 3-year observation window (Rußwurm and Körner, 2020; Zerveas et al., 2020).

For each target year, three consecutive years of Landsat observations are concatenated, encompassing the target year and the two immediately preceding years (year-2, year-1, year), and aggregated into a single annual cycle to densify the sparse annual time series. After aggregation, observations are chronologically sorted and padded to a fixed sequence length of 200 time steps. Positions corresponding to padded time steps receive a key-padding mask that prevents these positions from contributing to attention score computations (Vaswani et al., 2023). The actual acquisition day-of-year for each observation is encoded as a sinusoidal positional encoding, enabling the model to correctly represent the irregular temporal spacing of Landsat acquisitions and to distinguish phenological phases across the observation window without requiring interpolation to a regular grid (Yuan Yuan et al., 2020). Input features are normalised per band using a robust standardisation based on the median and interquartile range (quantiles q=0.1 and q=0.9) — with transform parameters fitted on the training data and frozen at inference time to prevent data leakage. A sensor platform embedding with $sensordim = 5$ encodes a multi-hot vector per year indicating which Landsat satellites (L4, L5, L7, L8, L9) contributed to the observations. This embedding is added to the pre-attention temporal encoding via a learned linear projection and makes the Transformer aware of inter-sensor radiometric and observation density differences over the 40-year archive. A post-attention pooling layer using a learned query vector collapses the temporal dimension into one single summary vector per output target, letting each targets learned query decide, which valid timesteps matter most for its prediction. In a setting with target classes from a wide range of phenological and spectral characteristics, this pooling strategy allows the model to learn class-specific temporal patterns across the observation window. The model heads are two-layer Multi-Layer-Perceptrons including Dropout and GeLU activation functions (Hendrycks and Gimpel, 2016). The classification model heads map the Transformer's output to class probabilities over eleven dominant tree species and three functional leaf type classes trained with two separate Cross Entropy Loss (CELoss) computations summed into a single loss value (dominant species and functional leaf type). With the dual loss the model learns to simultaneously capture species identity without losing the ability to capture the distinction between EVE and CON leaf types. The regression heads trained with Mean Squared Error Loss (MSELoss) maps the output to continuous EVE, DEC, and CON cover fractions.

## 2.4 Training procedure

The model is trained via a three-stage progressive workflow designed to maximise temporal stability and to accommodate the heterogeneous quality and temporal coverage of available reference data (Fig. 3).

Stage 1 - Contrastive pretraining. The backbone is trained using a joint loss that simultaneously optimises a supervised classification objective (DTS or LTC) using CELoss/MSELoss and an unsupervised normalised temperature-scaled cross-entropy contrastive loss (NT-Xent) (Chen et al., 2020): $L = \lambda\cdot$ NT-Xent + CE. For the contrastive term, positive pairs are formed from the same forest plot observed in two independently sampled three-year windows ($year_0$ and $year_1$ drawn at random), representing different but ecologically equivalent observation contexts for the same location. NT-Xent pushes the backbone representations of these positive pairs toward each other in embedding space while repelling representations of observations from different plots. This objective forces the backbone to learn year-invariant spectral-temporal features that are not confounded by observation density or inter-annual radiometric variability (Mañas et al., 2021). Positive pairs are further diversified by time series augmentation, which applies three independent perturbations to each view: random temporal masking (up to 20% of valid time steps zeroed), spectral noise injection (additive Gaussian noise with $\sigma = 0.01$), and band masking (10% of spectral bands zeroed per sample) (Yuan et al., 2025). For contrastive pretraining only data from csCFI and csCOVER are used, as these pure forest stands are considered stable over time. The contrastive and classification signals are routed through separate heads attached to the shared backbone: a two-layer MLP projection head for NT-Xent, and the classification/regression head for CELoss or MSELoss, respectively. The model is optimised using AdamW with a cosine learning rate schedule preceded by a 10% warm-up phase. Stage 1 training runs for 60 epochs with a batch size of 128. Reference data consist of csCFI, rsCFI, csCOVER, and rsCOVER. Class imbalance is addressed jointly by class-weighted CrossEntropyLoss and by sample weights set to the square root of the inverse class frequency, following the approach of Grabska-Szwagrzyk et al. (2024). A stratified 20% validation hold-out, stratified by plot polygon identifier, is maintained across all training stages to prevent spatial autocorrelation between training and validation sets.

Stage 2. Fine-tuning on labelled data. The contrastive backbone from Stage 1 is loaded and fine-tuned under a purely supervised objective (CrossEntropyLoss + MSELoss). The embedding layer is kept frozen in this stage to preserve the contrastive representations, while the task heads are retrained using a reduced learning rate to prevent catastrophic forgetting (Hiebl et al., 2025). This stage runs for 10 epochs on the full 1985–2025 Landsat time series, allowing the model to adapt the task heads to the full historical archive without disturbing the temporally invariant representations established during contrastive pretraining. The used reference data are the VDBI historical plot observations.

Stage 3. Fine-tuning on plot observation data. Additional fine-tuning passes are applied from the Stage 2 checkpoint giving priority to recent high-quality VPO field observations that provide the most accurate ground reference data for the present-day forest state. This stage is particularly intended to anchor the model's predictions of recent (post-2020) DTS and LTC to field-validated reference values, while the contrastive representations established in earlier stages ensure that this anchoring remains consistent with historical predictions.

Five independent model instances, differing in random seed, architecture micro-choices within TSTpad, and random subsets of training and validation data, are trained per stage. The five-seed ensemble delivers two outputs per pixel per year: the ensemble mean prediction (used as the primary data product) and the inter-seed standard deviation (used as the pixel-level uncertainty estimate). High inter-seed uncertainty identifies pixels where the model predictions are unstable, typically in regions of low observation density, unusual spectral composition, or high class confusion.

## 2.5 Mapping scheme

The forest prediction domain is defined by the ESA WordCover forested area (see 2.2.6). For each target year from 1985 to 2025, the corresponding three-year Landsat observation window (year-2 to year, inclusive) is assembled from the Landsat Collection-2 Level-2 archive, cloud-masked, and processed to the 14 spectral features described in Section 2.2.5. Features are normalised with the fixed training-time standardization parameters and padded to 100 time steps. The five trained ensemble models are applied independently to each forested pixel, and the resulting class probability distributions and cover fraction predictions are aggregated across ensemble members to produce the mean prediction and inter-member uncertainty layers.

Observation density varies substantially across the archive. Before 1999, when only Landsat 5 TM was operational, the

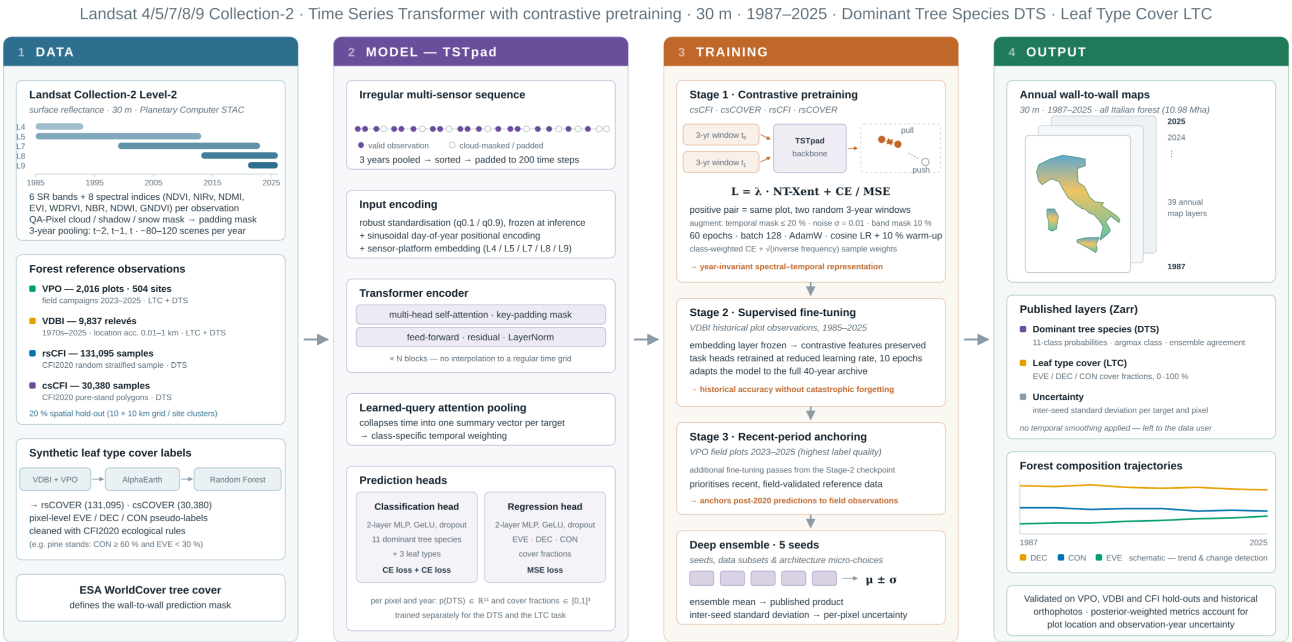


**Figure 3.** Workflow diagram illustrating the three-stage progressive training approach for forest mapping using Landsat time series data.

three-year window typically yields fewer cloud-free observations than the post-2013 period when Landsat 8 and 9 operate simultaneously. This density gradient is an analogue of the observational sampling bias documented for NDVI trend analyses by Bayle et al. (2024): predictions in the early archive period carry higher uncertainty due to sparse temporal sampling, which may cause the model to have lower confidence in phenological-season attribution.

We did not process the resulting annual time series using outlier detection or temporal smoothing, before storing the published LTC and DTS prediction maps. This was a design choice so downstream users can - and are encouraged to - apply temporal smoothing or outlier detection methods specifically appropriate to their application context. This is a necessary step for downstream tasks, as the time series contain both true ecological change, but also spurious noise from observation gaps, cloud contamination, and inter-annual radiometric variability.

## 2.6 Evaluation

The validation strategy is designed to assess both the accuracy of the map product at recent time steps and the temporal consistency of predictions across the full 40-year archive. Three independent validation datasets are employed. The VDBI hold-out test split provides the primary historical validation reference, with plot observations dating back to approximately 1995. This split allows accuracy assessment at historical time steps (1995–2025) where field data are available, spanning the Landsat 5-only and Landsat 5/7 overlap periods. The VPO hold-out split — collected during the 2023 to 2025 field work — provides the primary validation reference for recent predictions (post-2020) from the high-density Landsat 8/9 period. This dataset contains data from many mixed forest stands with hard-to-predict labels. The two validation datasets provide complementary temporal coverage: the VDBI split quantifies historical accuracy under lower observation density, while the VPO split quantifies accuracy under the best available observation conditions. A third validation source consists of a set of pure-stand forest polygons manually interpreted from high-resolution Google Earth Pro imagery in combination with the CFI dataset at recent time steps. These polygons, selected to represent spectrally unambiguous mono-specific stands, enable validation of the classification output under conditions where the expected dominant species label is highly reliable, providing an upper bound on classification accuracy unconstrained by label uncertainty in mixed stands or uncertain plot observation coordinates.

For the classification head, overall accuracy (OA) and per-class F1 score are computed for each period. For the regression heads, root mean squared error (RMSE) and mean absolute error (MAE) are reported for each of the three cover fractions (EVE, DEC, CON). The VPO dataset is finally used to assess in detail the model's ability to predict DTS and LTC in mixed stands, where the prediction of a single label is inherently more difficult and uncertain.

To account for uncertainties in plot observations coming from (i) a spatial mismatch between plot location and Landsat pixel, and (ii) temporal uncertainty and variations between plot observation year and Landsat acquisition year, we complement the direct accuracy metric with a true-label-conditioned metric (Lobo et al., 2020). Predictions are evaluated over a space-time neighborhood around each plots reported coordinates and year, weighting candidate locations by a Gaussian space-time prior (location and temporal uncertainty) combined with a Gaussian likelihood that downweights candidates whose predicted value strongly disagrees with the plots true label, following the logic of likelihood-based geolocation correction (Shannon et al., 2024). This estimator is bracketed by two limiting cases obtained by varying the likelihood bandwidth: a pure prior-weighted neighborhood mean that ignores the label entirely, and a best-match estimate that selects the single most consistent candidate location. We report accuracy metrics (RMSE, MAE and $R^2$) computed under all three methods to characterize how sensitive model accuracy is to possible location and temporal errors (see Appendix A).

## 3 Results and Discussion

The final maps provide following variables per pixel and per year (1987 to 2025):

- Dominant tree species: ensemble mean of dominant tree species class and level of agreement across ensemble members; ensemble mean of class probabilities and ensemble standard deviation for 11 classes
- Leaf type cover fractions: ensemble mean of EVE, DEC, and CON cover fractions and ensemble standard deviation

### 3.1 Performance metrics on test datasets

#### 3.1.1 Leaf type cover regression

The hold-out splits of the recent, methodologically homogeneous VPO dataset and of the multi-temporal VDBI collection are used to quantify the error of the annual LTC maps (Tab. 2). VPO plots were located with an accuracy that is high relative to the $30\,\mathrm{m}$ Landsat grid, so the estimator evaluated directly at the reported plot coordinate is the appropriate method for this dataset, while for VDBI, the combination of the three estimators can give insights into the sensitivity of the metrics to residual location and timing error (Sec. 2.6; see also Appendix A).

Evaluated directly at plot location, the maps reproduce DEC cover with an MAE of $9.7\,\%_{cover}$ (RMSE $12.4\,\%_{cover}$, $R^2 = 0.86$) and EVE cover with an MAE of $12.5\,\%_{cover}$ (RMSE $19.1\,\%_{cover}$, $R^2 = 0.66$). The consistently weaker performance for EVE has two plausible explanations. First, the ratio of RMSE to MAE is larger for EVE ($6.6\,\%_{cover}$) than for DEC ($2.7\,\%_{cover}$), indicating that the EVE error is driven by a comparatively small number of large residuals rather than by a uniform offset, i.e. by a subset of stands that is systematically misinterpreted rather than by generally erroneous predictions. Second, VPO reference cover is recorded separately for three vertical strata and aggregated across them, whereas Landsat observes a top-of-canopy signal. Parts of Italian EVE species frequently occur as an understory component beneath a deciduous overstory (Alessi et al., 2019; Conedera et al., 2018), a setup that is only weakly reflected in canopy spectral signal outside the leaf-off window (Chastain and Townsend, 2007; Yang et al., 2023). Reference EVE cover can therefore exceed the fraction of the signal that is physically available to the sensor, which inflates the model error.

CON cover has the lowest absolute errors on the VPO split (MAE $5.0\,\%_{cover}$, RMSE $9.1\,\%_{cover}$) but a negative coefficient of determination ($R^2 = -0.05$). This combination is a characteristic of the samples in the dataset rather than the models predictions. The VPO field data contain almost no CON-dominated stands, which means the CON variance is very small as all references and predictions are close to $5.0\,\%_{cover}$.

On the VDBI split the errors increase substantially for all three LTC fractions. Evaluated directly at the reported plot coordinate, the MAE roughly doubles relative to VPO — from 12.5 to $22.8\,\%_{cover}$ for EVE, from 9.7 to $20.7\,\%_{cover}$ for DEC, and from 5.0 to $9.8\,\%_{cover}$ for CON. RMSE increases in the same proportion (from 19.1 to 32.2, from 12.4 to 27.2, and from 9.1 to $18.0\,\%_{cover}$). Explained variance drops accordingly, from $R^2 = 0.66$ to 0.29 for EVE and from $R^2 = 0.86$ to 0.50 for DEC. CON is the only fraction that improves on VDBI ($R^2$ from $-0.05$ to 0.66), which confirms the interpretation given above: VDBI covers the alpine and Apennine conifer belts and therefore contains the reference variance that the VPO dataset lacks, so $R^2$ is an informative statistic for this fraction only on VDBI. This dataset aggregates plots collected over five decades by different observers, many of them without or with low-accuracy GNSS positioning, with reported location accuracies ranging from 0.01 to $1\,\mathrm{km}$ (Sec. 2.2.1). Positioning errors of this magnitude propagate directly into the accuracy m etrics, as field plot positioning error is known to bias remote sensing accuracy assessments negatively (Persson et al., 2022). Even under closed conifer canopy, GNSS positioning commonly deviates by several metres to tens of metres (Strunk et al., 2025), so a reported VDBI coordinate may address a Landsat pixel that is spectrally unrelated to the described stand. The VDBI figures evaluated directly at the reported coordinate are consequently not a clean estimate of model error, and we report them as a conservative lower bound on historical map quality rather than as an accuracy statement. Under the posterior-weighted estimator, which accounts for the location accuracy reported for each individual plot, the VDBI errors fall back to within 1.4 to $5.7\,\%_{cover}$ MAE of the VPO figures (18.2, 12.6 and $6.4\,\%_{cover}$ for EVE, DEC and CON, at $R^2 = 0.52$, 0.82 and 0.88), Which gives a more optimistic estimate of the model error.

The three estimator variants limit this effect quantitatively. For EVE on the VPO split, MAE decreases from 12.5 to 11.6

and further to $8.5\,\%_{cover}$, and $R^2$ increases from $0.66$ to $0.72$ and $0.79$, when moving from the direct estimator through the importance-weighted posterior to the best-match candidate. The width of this bracket differs markedly between the two datasets, and it scales with their reported coordinate accuracy. For DEC, the direct and best-match MAE differ by $1.3\,\%_{cover}$ on VPO (9.7 versus 8.4), but by $14.1\,\%_{cover}$ on VDBI (20.7 versus 6.6). We see the same contrast for the other two LTC fractions and for $R^2$, which is almost unchanged by posterior weighting on VPO (0.86 to 0.85 for DEC) but rises from $0.50$ to $0.82$ on VDBI. A roughly tenfold larger span on the dataset with metre- to kilometre-scale coordinate uncertainty is what would be expected if reference geolocation, rather than model error, dominated the VDBI mtrics. The gap between the direct and the best-match estimate is an upper bound on the fraction of the reported error that can be attributed to space-time reference mismatch rather than to the model, but it must not be read as a corrected accuracy value: the best-match estimator selects the neighbourhood candidate that agrees most closely with the reference label and is therefore optimistically biased, in the same way that label-informed geolocation correction needs an independent high-accuracy reference dataset to bias correct (Shannon et al., 2024). The true accuracy lies between the direct and the best-match figures. We therefore use the direct estimate as the reference value for VPO, whose coordinates are accurate relative to the Landsat grid, and the posterior-weighted estimate for VDBI, where the reported per-plot location accuracy contributes highly to the reported error (Tab. 2).

### 3.1.2 Dominant tree species classification

DTS classification performance is evaluated on the VPO, VDBI, and CFI test datasets (Tab. 3). The model reaches a weighted-average F1-score of $0.62$ on the VPO split (macro F1 $0.65$ over the six classes represented), $0.64$ on the VDBI split (macro F1 $0.60$), and $0.72$ on the CFI test dataset (macro F1 $0.71$ over all eleven classes). Overall accuracy is $0.62$ on both plot-based splits (VPO, $n = 493$; VDBI, $n = 536$) and $0.72$ on the considerably larger CFI test dataset ($n = 6642$). The gap of ten percentage points between the plot-based splits and the polygon-based CFI evaluation reflects the differing label quality, plot location accuracy but also the mixed stand character of VPO and VDBI datasets compared to CFI dataset.

The per-class results are markedly asymmetric between precision and recall. On the VPO split beech is recovered with high recall but low precision (0.91 / 0.57), whereas deciduous oak shows the inverse pattern (0.46 / 0.88). The model assigns beech to a substantial share of oak-dominated plots, but the predicted beech plots are mostly correct. The direction of this asymmetry is not stable across the three evaluations, however. On VDBI and CFI beech behaves in the opposite way and is the more conservatively assigned class (precision $0.88$ and $0.83$ at recall $0.71$ and $0.65$), so the overprediction of beech is a property of the VPO plot sample, dominated by mixed central and southern Apennine stands, and not a general property of the model. This highlights the discrepancy of labelling labelling in different datasets. What is common to all three evaluations is the confusion of spectrally and phenologically similar temperate broadleaves, which is well documented for national-scale Sentinel-2 mapping (Grabska-Szwagrzyk et al., 2024; Hemmerling et al., 2021). The manna ash–hornbeam class (*Ornus–Ostrya*) exhibits the lowest accuracy metrics across all three datasets (F1 0.43–0.63). These stands are typically structurally heterogeneous and successional, so a single dominant-species label is a bad characterisation of the pixel. The aggregate “other deciduous”, “other coniferous” and “other evergreen” classes behave as expected for categories, that are supposed to catch residual classes: on the VDBI split, “other deciduous” reaches a recall of $0.81$ at a precision of $0.23$, which means it absorbs plots that the model cannot attribute to any specific class.

Per-class metrics based on small test supports require particular caution. The highest F1-score in the VPO evaluation (chestnut, $0.92$) rests on $26$ test plots, and the spruce metrics in the VDBI evaluation on $15$ plots; at these sample sizes the confidence intervals are wide enough that the values cannot be interpreted as class-level accuracy statements. The challenges of estimating reliable per-class accuracies for infrequent classes under high class imbalance is a common limitation of large-area species mapping (Hemmerling et al., 2021; Grabska-Szwagrzyk et al., 2024). Five of the eleven classes (silver fir, spruce, pine, other coniferous and other evergreen) are not represented in the VPO split at all, because the 2023–2025 campaigns targeted protected areas in central and southern Italy. VPO therefore constrains the recent-period accuracy of the broad-leaved classes only, and the alpine conifer classes are evaluated exclusively through VDBI and CFI.

The confusion matrix for the CFI test dataset (Fig. 4) shows which class pairs carry the residual error. Misclassification is overwhelmingly constrained to the functional leaf type of the reference class. Of the $1877$ misclassified plots, $1386$ ($74\,\%$) are assigned to another class of the same leaf type group, so the correct leaf type is predicted for $92.6\,\%$ of all test plots while the species label is correct for only $72\,\%$. The DTS layer is therefore substantially more reliable when aggregated to leaf type than when read at species level, which is consistent with the LTC product being the more robust of the two data layers.

The weakest class is “other coniferous” (recall $0.50$, precision $0.49$). Only $109$ of its $220$ plots are predicted correctly, $43$ ($20\,\%$) are predicted as pine, which is defensible for a class that contains minor and largely Mediterranean coniferous stands, but $66$ plots ($30\,\%$) are assigned to “other deciduous” and to deciduous oak. This is the only class in which the majority of the error crosses the leaf type boundary, indicating that the “other coniferous” category is heterogeneous and does not have a consistent spectral–temporal signature. Among the deciduous classes, deciduous oak has the lowest recall (0.61) despite the largest support ($n = 1137$): 142 plots (12 %) are predicted as chestnut, 134 (12 %) as manna ash–hornbeam and 90 (8 %) as evergreen oak, the two first having the largest off-diagonal counts in the confusion matrix. The reverse direction is equally pronounced, with 117 manna ash–hornbeam plots (17 %) predicted as chestnut and 123 evergreen oak plots (11 %) predicted as deciduous oak. Deciduous oak, chestnut and manna ash–hornbeam therefor form three confusion pairs. the three co-occur across the sub-Mediterranean and lower montane belt and frequently form mixed stands in which a single dominant-species label might only approximately describe a 30 m pixel (Bricca et al., 2026).

The deciduous oak–evergreen oak confusion (90 and 123 plots in the two directions) is the largest across-leaf-type confusion in the matrix and deserves separate attention, because it affects the transition that is of primary ecological interest. These are the stands in which evergreen and deciduous oaks co-occur and in which evergreen broad-leaved species happen to expand (Alessi et al., 2019; Conedera et al., 2018). Trajectories in this class pair should be interpreted together with the continuous LTC layers and the ensemble standard deviation rather than from the argmax class alone. Manna ash–hornbeam and beech both reach a recall of 0.65. Beech errors are distributed almost evenly over the four remaining deciduous classes (6–10 % each) without a dominant confusion pair, whereas manna ash–hornbeam loses 17 % of its plots to chestnut alone while absorbing 287 plots from other classes (134 deciduous oak, 56 beech, 41 chestnut and 39 other evergreen). The most extreme sink behaviour is shown by “other deciduous”. It recovers 85 % of its own 240 plots but additionally receives 351 plots from seven other classes — more than its entire support — which is why its precision collapses to 0.37 at a recall of 0.85.

Within the coniferous block the metrics are more favourable. Spruce reaches a recall of 0.72, and all 102 of its misclassified plots remain within the coniferous group (54 silver fir, 35 other coniferous, 13 pine); its precision of 0.95 is the highest of any class. Silver fir attains the highest recall among the species-level classes (0.85, with 7 % loss to pine and 5 % to beech) and pine reaches 0.72. Conifer confusion is therefore predominantly a problem between structurally and phenologically similar evergreen needle-leaved stands. Finally, the CFI matrix is computed on photo-interpreted CFI2020 reference polygons under the posterior-weighted estimator, i.e. under the most favourable label conditions available in this study. The confusions described above are those that persist even where the reference label is unambiguous, and they must be expected to be more severe in the mixed stands that dominate the mapped area (Blickensdörfer et al., 2024).

#### 3.1.3 Pure versus mixed stands

#### 3.1.4 Comparison with published products and scope of the reported metrics

The LTC accuracies are broadly comparable to published fractional composition products, although the comparison is only an indicator. Direct prediction of tree species proportions from Sentinel-2 with a spectral–spatial CNN in Belgium achieved an adjusted $R^2$ of 0.50 for the predicted proportions and an overall accuracy of 0.73 for the majority species (Bolyn et al., 2022), and the pan-European deciduous–coniferous proportion product derived from Sentinel-2 and harmonised national forest inventory plots reports $R^2$ values of 0.52–0.73 with relative RMSE between 30 and 65 % across processing areas (Miettinen et al., 2025). The DEC results reported here ($R^2 = 0.86$) fall above and the EVE results ($R^2 = 0.66$) within that range. But several differences have to be pointed out for the comparison. The reference quantity differs (projected cover per leaf type in this study versus basal-area or biomass proportions in the cited products), the sensor and grid differ (30 m Landsat versus 10 m Sentinel-2), the class definitions are coarser here, and the evaluation samples differ in spatial design.

### 3.2 Dominant tree species dynamics

TODO. - figure of transitions of 3 most likely transformations - figure of biggest areas of change - table per class of highest losses/gains

### 3.3 Leaf type cover dynamics

TODO. - figure of theil sen slopes of the 3 target classes for all of italy - table of forest type/habitat type with highest changes

### 3.4 Limitations and future work

The reportedaccuracy of this dataset is limited more by the reference data than by the model predictions. Reported VDBI location accuracies span two orders of magnitude (Sec. 2.2.1), and a coordinate displaced by more than one pixel attaches its label to an unrelated stand, degrading both training and the reported metrics. Receivers require long occupations to reach metre-level accuracy under closed canopy (Strunk et al., 2025). Such reference error biases accuracy assessments negatively (Persson et al., 2022; Elmes et al., 2020), which is why the VDBI figures in Tab. 2 are bracketed by three estimators rather than reported as a single value. Although the brackets can not be seen as a correction: likelihood-based geolocation correction is unbiased only against an independent high-accuracy reference (Shannon et al., 2024), which does not exist for the historical period. The VDBI-based figures should therefore be read as

**Table 2.** Regression performance (MAE, RMSE, $R^2$) for leaf type cover (LTC) prediction, evaluated on two test datasets. X/X/X refers to the computation of the evaluation metrics, being directly from plot location, importance-weighting, and best-match (see Sec. 2.6 and Appendix A). MAE and RMSE are reported as $\%_{cover}$.

| | VPO test split | | | VDBI test split | | |
|---|---|---|---|---|---|---|
| target | MAE | RMSE | $R^2$ | MAE | RMSE | $R^2$ |
| EVE | **12.5** / 11.6 / 8.5 | **19.1** / 17.5 / 14.9 | **0.66** / 0.72 / 0.79 | 22.8 / **18.2** /10.9 | 32.2 / **26.3** / 21.3 | 0.29 / **0.52** / 0.69 |
| DEC | **9.7** / 8.7 / 8.4 | **12.4** / 10.6 / 7.9 | **0.86** / 0.85 / 0.92 | 20.7 / **12.6** / 6.6 | 27.2 / **16.1** / 11.9 | 0.50 / **0.82** / 0.90 |
| CON | 5.0 / 4.4 / 3.0 | 9.1 / 7.4 / 6.1 | -0.05 / 0.31 / 0.54 | 9.8 / **6.4** / 3.2 | 18.0 / **11.1** / 8.7 | 0.66 / **0.88** / 0.92 |

**Table 3.** Classification performance (precision, recall, f1-score) for the dominant tree species classification task, evaluated on three test datasets. Reported is the likelihood-based importance-weighted performance result based on the location accuracy of the plots (see Sec. 2.6).

| | VPO test split | | | | VDBI test split | | | | CFI test dataset | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| class | prec. | recall | f1-score | support | prec. | recall | f1-score | support | pre. | recall | f1-score | support |
| silver fir | - | - | - | - | - | - | - | - | 0.86 | 0.85 | 0.86 | 518 |
| other dec. | 0.52 | 0.55 | 0.53 | 112 | 0.23 | 0.81 | 0.36 | 27 | 0.37 | 0.85 | 0.52 | 240 |
| dec. oak | 0.88 | 0.46 | 0.60 | 127 | 0.71 | 0.40 | 0.51 | 123 | 0.67 | 0.61 | 0.64 | 1137 |
| chestnut | 0.96 | 0.88 | 0.92 | 26 | 0.89 | 0.62 | 0.73 | 30 | 0.67 | 0.74 | 0.70 | 817 |
| other con. | - | - | - | - | - | - | - | - | 0.49 | 0.50 | 0.49 | 220 |
| beech | 0.57 | 0.91 | 0.68 | 108 | 0.88 | 0.71 | 0.84 | 120 | 0.83 | 0.65 | 0.73 | 594 |
| eve. oak | 0.65 | 0.64 | 0.65 | 53 | 0.69 | 0.71 | 0.70 | 110 | 0.86 | 0.79 | 0.82 | 1141 |
| other eve. | - | - | - | - | - | - | - | - | 0.89 | 0.81 | 0.85 | 503 |
| pine | - | - | - | - | 0.90 | 0.82 | 0.86 | 51 | 0.74 | 0.72 | 0.73 | 410 |
| ornus-ostr. | 0.69 | 0.43 | 0.53 | 51 | 0.40 | 0.50 | 0.45 | 60 | 0.61 | 0.65 | 0.63 | 695 |
| spruce | - | - | - | - | 0.78 | 0.47 | 0.58 | 15 | 0.95 | 0.72 | 0.82 | 367 |
| accuracy | | | 0.62 | 493 | | | 0.62 | 536 | | | 0.72 | 6642 |
| macro avg | 0.71 | 0.65 | 0.65 | | 0.67 | 0.61 | 0.60 | | 0.72 | 0.72 | 0.71 | |
| weighted avg | 0.68 | 0.62 | 0.62 | | 0.71 | 0.62 | 0.64 | | 0.75 | 0.72 | 0.72 | |

bounds and the VPO-based figures as the operationally relevant accuracy.

No dataset available for Italy allows the mapped trajectories to be validated directly. VDBI is the only reference source distributed across the mapped time period, and its plots are clustered in well-studied vegetation types and regions, a bias common to European vegetation-plot archives (Chytrý et al., 2016; Sabatini et al., 2021). Reported accuracies therefore describe per-pixel performance at plot locations and are not area-adjusted map accuracies (Olofsson et al., 2014); users requiring the latter for a specific region or period should conduct a probability-based assessment on the published maps. Resurveys of historical vegetation plots (Kapfer et al., 2017; Verheyen et al., 2017; Midolo et al., 2026) and the reorientation of the Italian national inventory towards continuous monitoring (D'Amico et al., 2025) are the two necessary steps to a full time trajectory validation in future studies.

Annual predictions are produced independently, with no temporal model linking individual years. Three-year pooling smoothes but does not remove inter-annual variability caused by changing observation density, cloud contamination or inter-annual phenological differences, so DTS pixels may oscillate between spectrally similar species in consecutive years even where the stand is actually stable. We publish unsmoothed and unprocessed time series predictions (Sec. 2.5), but users are encouraged to apply andvanced methods to stabilize the time trajectories. E.g.,a hidden Markov formulation with per-year ensemble probabilities as emissions, an ecologically constrained transition matrix, and Viterbi smoothing, could suppress such flips while preserving true class transitions (Abercrombie and Friedl, 2016; Hermosilla et al., 2018), especially if combined with spectral trajectories (Kennedy et al., 2010; Zhu and Woodcock, 2014) or independently mapped disturbance dates (Viana-Soto and Senf, 2025). This would be a promising improvement to the DTS layer, which could also be applied to the LTC layer in an adapted way.

The input archive itself remains heterogeneous. Usable satellite observations per pixel and year increase from the Landsat 5 period to present (Bayle et al., 2024; White et al., 2014), SLC-off gaps in Landsat-7 observations affect all years from 2003 onwards, and radiometric differences between the different sensors aboard the Landsat missions (TM,

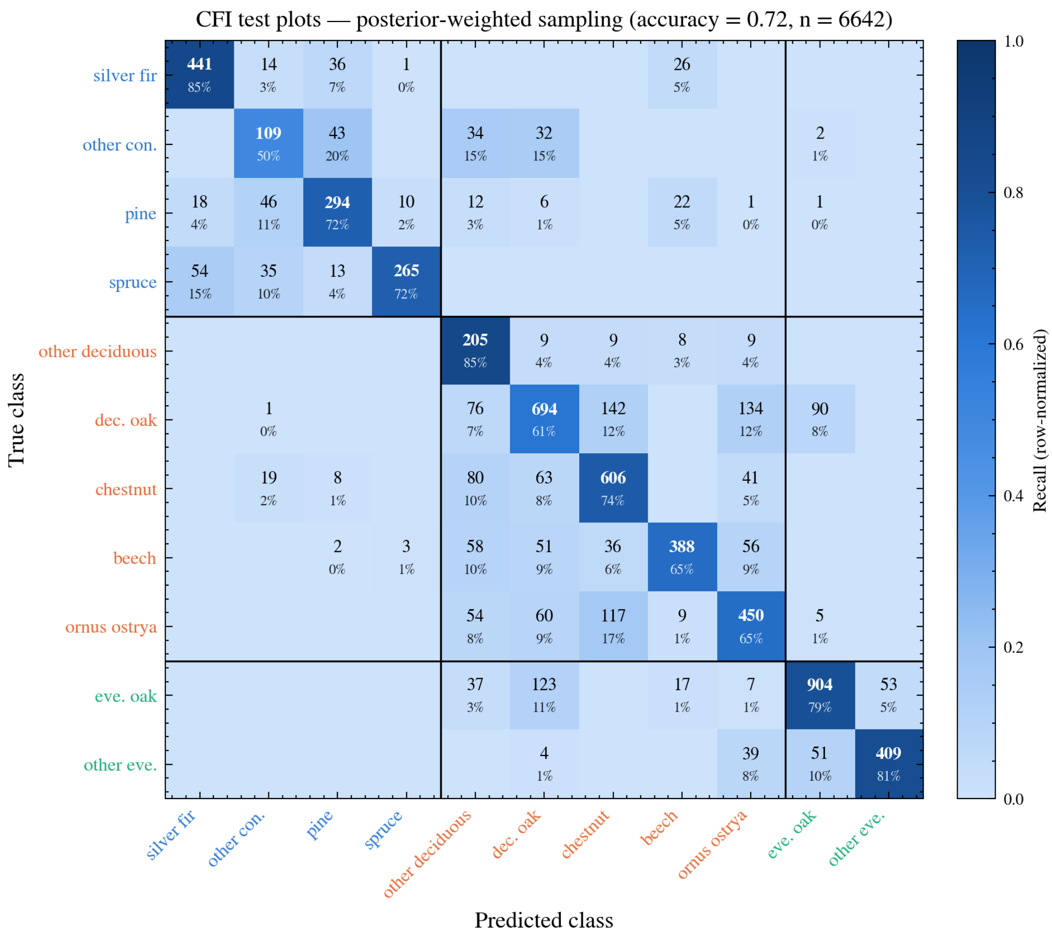


**Figure 4.** Row-normalised confusion matrix (recall) for the dominant tree species classification on the CFI test dataset ($n = 6642$), computed with the posterior-weighted estimator (Sec. 2.6). Cell labels give the absolute number of test plots and the row-normalised share. Classes are ordered and grouped by functional leaf type (coniferous, deciduous broad-leaved, evergreen broad-leaved); black lines delineate the three groups, so that off-diagonal cells outside the diagonal blocks are confusions that also change the predicted leaf type.

ETM+ and OLI) persist after Level-2 processing (Roy et al., 2016). The sensor embedding and three-year pooling make the model aware of these effects but cannot separate true vegetation change from artefacts of changing observation conditions. A cross-calibrated, gridded full-archive product in the sense of Landsat Analysis Ready Data (Dwyer et al., 2018), FORCE (Frantz, 2019) or the Harmonized-Landsat-Sentinel-2 archive (Claverie et al., 2018) would remove much of this uncertainty. In the current case the pre-1999 maps have the highest uncertainty of the record and all maps should be used together with the ensemble standard deviation layers.

Mapping cost is a practical constraint on how often the product can be updated. Although parallelized, fetching, masking and featurising three years of observations per target year for every forested pixel and then running five ensemble members required several weeks of continuous processing, dominated by data transfer and preprocessing, and finally model inference (Gorelick et al., 2017; Frantz, 2019). Pretrained representations are the most promising route out of both this and the label constraint, since analysis-ready embeddings reduce a year of observations to a compact per-pixel vector consumable by lightweight models (Brown et al., 2025; Feng et al., 2026; Ball et al., 2026; Hiebl et al., 2026). The synthetic LTC labels used here were generated this way (Sec. 2.2.4). Existing embedding products, however, cover only the modern multi-sensor era. A Landsat-native backbone pretrained on unlabelled pixel time series (Tseng et al., 2024; Stewart et al., 2025) would extend this benefit to the full available Landsat archive and reduce each new mapping task to a fine-tuning step.

## 4 Conclusions

The Landsat collections provide an unprecedented archive of satellite imagery for reconstructing and hindcasting environemntal variables across large regions. Few studies exist, that exploit the full 40 years of available Collection-2 imagery to map forest related variables and provide time series datasets, that can be used to investigate forest dynamics and changes under climate and landuse change conditions. This study predicts annual maps of dominant leaf type and leaf type cover from Landsat time series, forest plot observations and forest inventories.

## 5 Code and data availability

All python code that was used during data processing, training and mapping is available under https://git.uibk.ac.at/c7161037/ls_mapping (training) and https://git.uibk.ac.at/rslab/sattstools (beta version, remote sensing data processing). The model architecture and model weights can be found on "link text". The Zarr-compressed annual maps of leaf type cover fractions and dominant tree species probabilities are available on "link text".

## Appendix A: Computation of evaluation metrics

All accuracy figures reported in Sect. 2.6 are computed from the published maps, this means the estimators below are applied to the same spatio-temporal product that is published, and not to model outputs held in memory. For a plot $p$ with reported coordinates $\mathbf{s}_p$ and observation year $t_p$, let $\mathcal{N}_p$ denote the set of candidate map pixels $(\mathbf{s},t)$ with $\|\mathbf{s}-\mathbf{s}_p\|_\infty \leq 2\sigma_s$ and $|t-t_p| \leq 2\sigma_t$, clipped to the temporal extent of the product. This creates a squared box of half-width $2/sigma_s$ around the plot coordinates. The *direct* estimator is the special case $\mathcal{N}_p = \{(\mathbf{s}_p, t_p)\}$, i.e. the nearest pixel of the reported year. The two estimators described below are conditioned on the reference plots label and are therefore validation diagnostics only, as they cannot be applied where the reference value is unknown. Pixels without a valid prediction receive zero weight and are excluded, and plot–prediction pairs containing a missing value are dropped before metrics are computed.

### A1 Posterior-weighted estimator

Every candidate cell in $\mathcal{N}_p$ is treated as a hypothesis for the true plot location and observation date. The prior expresses the reported position and year and their uncertainty as a separable Gaussian kernel,

$$w^{\text{prior}}(\mathbf{s},t) = \exp\left(-\frac{\|\mathbf{s}-\mathbf{s}_p\|^2}{2\sigma_s^2}\right)\exp\left(-\frac{(t-t_p)^2}{2\sigma_t^2}\right), \quad \text{(A1)}$$

with position uncertainty $\sigma_s =$ plot location accuracy in m and temporal uncertainty $\sigma_t = 1\,\text{a}$, clamped at $2\sigma_s$ and $2\sigma_t$. For the LTC task the likelihood of a hypothesis is the agreement between the mapped cover fraction $\hat{y}_k(\mathbf{s},t)$ of leaf type $k \in \{\text{EVE}, \text{DEC}, \text{CON}\}$ and the plot reference cover $y_{p,k}$,

$$w_k^{\text{lik}}(\mathbf{s},t) = \exp\left(-\frac{\left(\hat{y}_k(\mathbf{s},t)-y_{p,k}\right)^2}{2\sigma_{\text{obs},k}^2}\right), \quad \text{(A2)}$$

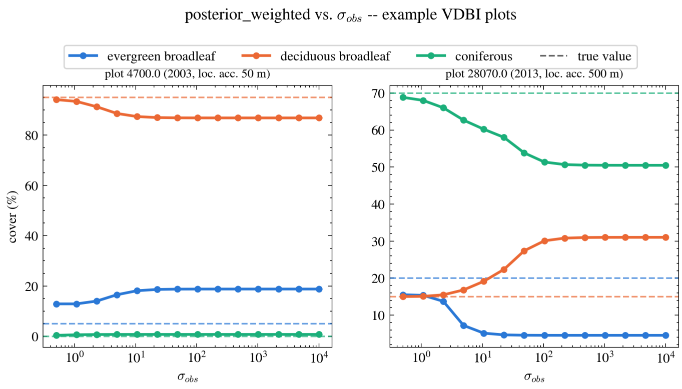


**Figure A1.** Sensitivity study of the observation bandwidth $\sigma_{\text{obs},k}$ that controls the reference-based likelihood landscape for the posterior-weighted estimator. With lower $\sigma_{\text{obs},k}$ (obsdervation window) the LTC estimate approaches the best-match and therefore true observation value (dashed lines).

and the estimate is the posterior-weighted mean over the neighbourhood,

$$\hat{y}_{p,k}^{\text{PW}} = \frac{\sum_{(\mathbf{s},t)\in\mathcal{N}_p} w^{\text{prior}} w_k^{\text{lik}}\, \hat{y}_k(\mathbf{s},t)}{\sum_{(\mathbf{s},t)\in\mathcal{N}_p} w^{\text{prior}} w_k^{\text{lik}}}. \quad \text{(A3)}$$

The observation bandwidth $\sigma_{\text{obs},k}$ is set per leaf type to the standard deviation of the residuals of the direct estimator on the VPO split, i.e. on the dataset with the lowest location uncertainty, so that the likelihood is not sharper than the residual scatter that is already explained by model error alone. It acts as a single dial between the two limiting cases: $\sigma_{\text{obs},k} \to \infty$ removes the label from the weighting and recovers a purely prior-weighted neighbourhood mean, whereas $\sigma_{\text{obs},k} \to 0$ recovers the best-match estimator of Sect. A2. This is similar to a sampling-importance-resampling implementation that draws $n$ candidates directly from the prior and weights them by the likelihood alone (Lobo et al., 2020).

For the DTS task the same weights are applied to the predicted class-probability vector $\hat{\mathbf{p}}(\mathbf{s},t)$. The likelihood of a hypothesis is the probability that the candidate cell assigns to the reference class.

Spatial uncertainty $\sigma_s$ is set for each plot individually, depicting the reported accuracy of the plot coordinates based on GNSS measurements or data curator expert knowledge. $\sigma_s$ ranges between $1\,\text{m}$ and $1000\,\text{m}$ in the VDBI dataset, which guides the prior distribution over the spatial dimension and the size of the neighbourhood (see Fig. A2). The sensitivity of the posterior-weighted estimator to the choice of $\sigma_{\text{obs},k}$ is illustrated in Fig. A1, and example posterior landscapes are shown in Fig. A2.

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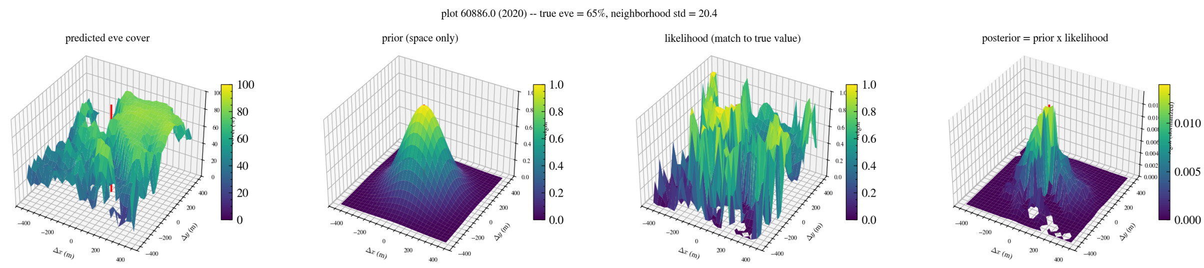


**Figure A2.** Posterior weight landscape for an example plot with highly heterogenous EVE cover within the neighbourhood computed from the gaussian prior and likelihood for the posterior-weighted estimator. The LTC estimate is weighted within the obsevation window defined by the prior distribution and the likelihood of the reference value, downweighting candidate pixels that are further away and specifically dissimilar to the reference value.

## A2 Best-match estimator

The best-match estimator replaces the weighted mean by a hard selection over a neighbourhood ($2\sigma_s$, plot location accuracy in $\mathrm{m}$, and a fixed window of $t_p \pm 1\,\mathrm{a}$). For the LTC task it returns, per leaf type, the value of the single candidate cell that agrees most closely with the reference,

$$\hat{y}_{p,k}^{\mathrm{BM}} = \hat{y}_k \left( \underset{(\mathbf{s},t)\in\mathcal{N}_p}{\arg\min} \left| \hat{y}_k(\mathbf{s},t) - y_{p,k} \right| \right), \tag{A4}$$

and for the DTS task the label of the candidate cell that assigns the highest probability to the reference class. Because the selection is made using the reference value itself, this estimator is optimistically biased by construction: any leaf-type value present anywhere in the neighbourhood can be matched, and a plot is counted as correctly classified whenever at least one candidate cell within the plot location accuracy reported by the data providers and $\pm 1\,\mathrm{a}$ carries the reference class. The resulting metrics are therefore not corrected accuracies but an upper optimistic bound on what could be attributed to spatial(-temporal) reference mismatch rather than to model error.

- figure: sensitivity study for sigmaobs in the posterior method. lower bound direct, upper bound best match, line in between with sigma
- figure of posterior landsacape for 2 example plots. prior (gaussian) * likelihood => posterior

## Appendix B: Comparison to historical orthophoto imagery

TODO.

### Author contributions.

TODO.

### Competing interests.

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### Disclaimer.

TODO.

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## References

Abercrombie, S. P. and Friedl, M. A.: Improving the Consistency of Multitemporal Land Cover Maps Using a Hidden Markov Model, IEEE Transactions on Geoscience and Remote Sensing, 54, 703–713, https://doi.org/10.1109/TGRS.2015.2463689, 2016.

Alessi, N., Těšitel, J., Zerbe, S., Spada, F., Agrillo, E., and Wellstein, C.: Ancient refugia and present-day habitat suitability of native laurophylls in Italy, Journal of Vegetation Science, 30, 564–574, https://doi.org/10.1111/jvs.12743, 2019.

Ball, J. G., Wicklein, J. A., Feng, Z., Knezevic, J., Jaffer, S., Madhavapeddy, A., Atzberger, C., Dalponte, M., and Coomes, D.: Geospatial foundation models enable data-efficient tree species mapping in temperate mountain forests, https://doi.org/10.64898/2026.02.23.707022, iSSN: 2692-8205 Pages: 2026.02.23.707022 Section: New Results, 2026.

Bayle, A., Gascoin, S., Berner, L. T., and Choler, P.: Landsat-based greening trends in alpine ecosystems are inflated by multidecadal increases in summer observations, Ecography, 2024, e07 394, https://doi.org/10.1111/ecog.07394, _eprint: https://nsojournals.onlinelibrary.wiley.com/doi/pdf/10.1111/ecog.07394, 2024.

Bell, D. M., Gregory, M. J., and Yang, Z.: Hindcasting and updating Landsat-based forest structure mapping across years to support forest management and planning, Forest Ecology and Management, 572, 122 239, https://doi.org/10.1016/j.foreco.2024.122239, 2024.

Blickensdörfer, L., Oehmichen, K., Pflugmacher, D., Kleinschmit, B., and Hostert, P.: National tree species mapping using Sentinel-1/2 time series and German National Forest Inventory data, Remote Sensing of Environment, 304, 114 069, https://doi.org/10.1016/j.rse.2024.114069, 2024.

Bolyn, C., Lejeune, P., Michez, A., and Latte, N.: Mapping tree species proportions from satellite imagery using spectral–spatial deep learning, Remote Sensing of Environment, 280, 113 205, https://doi.org/10.1016/j.rse.2022.113205, 2022.

Bricca, A., Zerbe, S., Sabatini, F. M., Hiebl, B., Rutzinger, M., di Musciano, M., Calvia, G., Chiarucci, A., Poschlod, P., Rossi, C., Alessi, N., and Bonari, G.: Topography and Soil Moisture Regulate the Temperature-Biodiversity Relationship of Forests, Global Ecology and Biogeography, 35, e70 186, https://doi.org/10.1111/geb.70186, _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/geb.70186, 2026.

Brown, C. F., Kazmierski, M. R., Pasquarella, V. J., Rucklidge, W. J., Samsikova, M., Zhang, C., Shelhamer, E., Lahera, E., Wiles, O., Ilyushchenko, S., Gorelick, N., Zhang, L. L., Alj, S., Schechter, E., Askay, S., Guinan, O., Moore, R., Boukouvalas, A., and Kohli, P.: AlphaEarth Foundations: An embedding field model for accurate and efficient global mapping from sparse label data, https://doi.org/10.48550/arXiv.2507.22291, arXiv:2507.22291 [cs], 2025.

Chastain, R. A. and Townsend, P. A.: Use of Landsat ETM and topographic data to characterize evergreen understory communities in Appalachian deciduous forests, Photogrammetric Engineering & Remote Sensing, 73, 563–575, 2007.

Chelli, S., Wellstein, C., Campetella, G., Canullo, R., Tonin, R., Zerbe, S., and Gerdol, R.: Climate change response of vegetation across climatic zones in Italy, Climate Research, 71, 249–262, https://doi.org/10.3354/cr01443, 2017.

Chen, T., Kornblith, S., Norouzi, M., and Hinton, G.: A Simple Framework for Contrastive Learning of Visual Representations, https://doi.org/10.48550/arXiv.2002.05709, arXiv:2002.05709 [cs], 2020.

Chytrý, M., Hennekens, S. M., Jiménez-Alfaro, B., Knollová, I., Dengler, J., Jansen, F., Landucci, F., Schaminée, J. H. J., Aćić, S., Agrillo, E., et al.: European Vegetation Archive (EVA): an integrated database of European vegetation plots, Applied Vegetation Science, 19, 173–180, https://doi.org/10.1111/avsc.12191, 2016.

Claverie, M., Ju, J., Masek, J. G., Dungan, J. L., Vermote, E. F., Roger, J.-C., Skakun, S. V., and Justice, C.: The Harmonized Landsat and Sentinel-2 surface reflectance data set, Remote Sensing of Environment, 219, 145–161, https://doi.org/10.1016/j.rse.2018.09.002, 2018.

Conedera, M., Wohlgemuth, T., Tanadini, M., and Pezzatti, G. B.: Drivers of broadleaved evergreen species spread into deciduous forests in the southern Swiss Alps, Reg Environ Change, 18, 425–436, https://doi.org/10.1007/s10113-017-1212-7, 2018.

D'Amico, G., Chirici, G., Campetella, G., Canullo, R., Cervellini, M., Chelli, S., Di Biase, R. M., Fattorini, L., Floris, A., Franceschi, S., Francini, S., Giannetti, F., Marcelli, A., Marcheselli, M., Mattioli, W., Notarangelo, M., Papitto, G., Parisi, F., Pisani, C., Saba, E. P., Quatrini, V., Vangi, E., and Corona, P.: National Forest Inventory in Italy: new perspectives for forest monitoring, Annals of Forest Science, 82, 35, https://doi.org/10.1186/s13595-025-01303-9, 2025.

Dwyer, J. L., Roy, D. P., Sauer, B., Jenkerson, C. B., Zhang, H. K., and Lymburner, L.: Analysis Ready Data: Enabling Analysis of the Landsat Archive, Remote Sensing, 10, 1363, https://doi.org/10.3390/rs10091363, 2018.

Elmes, A., Alemohammad, H., Avery, R., Caylor, K., Eastman, J. R., Fishgold, L., Friedl, M. A., Jain, M., Kohli, D., Laso Bayas, J. C., Lunga, D., McCarty, J. L., Pontius, R. G., Reinmann, A. B., Rogan, J., Song, L., Stoynova, H., Ye, S., Yi, Z.-F., and Estes, L.: Accounting for Training Data Error in Machine Learning Applied to Earth Observations, Remote Sensing, 12, 1034, https://doi.org/10.3390/rs12061034, 2020.

Feng, Z., Atzberger, C., Jaffer, S., Knezevic, J., Sormunen, S., Young, R., Lisaius, M. C., Immitzer, M., Jackson, T., Ball, J., Coomes, D. A., Madhavapeddy, A., Blake, A., and Keshav, S.: TESSERA: Temporal Embeddings of Surface Spectra for Earth Representation and Analysis, https://doi.org/10.48550/arXiv.2506.20380, arXiv:2506.20380 [cs.LG], 2026.

Frantz, D.: FORCE—Landsat + Sentinel-2 Analysis Ready Data and Beyond, Remote Sensing, 11, 1124, https://doi.org/10.3390/rs11091124, 2019.

Gasparini, P., Di Cosmo, L., Floris, A., and De Laurentis, D., eds.: Italian National Forest Inventory—Methods and Results of the Third Survey: Inventario Nazionale delle Foreste e dei Serbatoi Forestali di Carbonio—Metodi e Risultati della Terza Indagine, Springer Tracts in Civil Engineering, Springer International Publishing, Cham, ISBN 978-3-030-98677-3 978-3-030-98678-0, https://doi.org/10.1007/978-3-030-98678-0, 2022.

Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., and Moore, R.: Google Earth Engine: Planetary-scale geospatial analysis for everyone, Remote Sensing of Environment, 202, 18–27, https://doi.org/10.1016/j.rse.2017.06.031, 2017.

Grabska-Szwagrzyk, E., Tiede, D., Sudmanns, M., and Kozak, J.: Map of forest tree species for Poland based on Sentinel-2 data, Earth System Science Data, 16, 2877–2891, https://doi.org/10.5194/essd-16-2877-2024, 2024.

Grünig, M., Rammer, W., Senf, C., Albrich, K., André, F., Augustynczik, A. L. D., Baumann, M., Bohn, F. J., Bouwman, M., Bugmann, H., Collalti, A., Cristal, I., Dalmonech, D., De Coligny, F., Dobor, L., Dollinger, C., Espelta, J. M., Forrester, D. I., Garcia-Gonzalo, J., González-Olabarria, J. R., Hiltner, U., Hlásny, T., Honkaniemi, J., Huber, N., Jonard, M., Jönsson, A. M., Kunstler, G., Lagergren, F., Lindner, M., Mina, M., Moos, C., Morin, X., Muys, B., Nabuurs, G.-J., Nieberg, M., Patacca, M., Peltoniemi, M., Reyer, C. P. O., Schelhaas, M.-J., Storms, I., Thom, D., Toïgo, M., and Seidl, R.: Climate change will increase forest disturbances in Europe throughout the 21st century, Science, 391, eadx6329, https://doi.org/10.1126/science.adx6329, 2026.

Hemmerling, J., Pflugmacher, D., and Hostert, P.: Mapping temperate forest tree species using dense Sentinel-2 time series, Remote Sensing of Environment, 267, 112 743, https://doi.org/10.1016/j.rse.2021.112743, 2021.

Hendrycks, D. and Gimpel, K.: Gaussian Error Linear Units (GELUs), https://arxiv.org/abs/1606.08415v5, 2016.

Hermosilla, T., Wulder, M. A., White, J. C., Coops, N. C., and Hobart, G. W.: Disturbance-Informed Annual Land Cover Classification Maps of Canada's Forested Ecosystems for a 29-Year Landsat Time Series, Canadian Journal of Remote Sensing, 44, 67–87, https://doi.org/10.1080/07038992.2018.1437719, 2018.

Herraiz, A. D., Salazar-Zarzosa, P., Acosta-Muñoz, C., Hernández-Clemente, R., and Villar, R.: Aridity-induced phenological shifts and greening trends in Mediterranean forest species: Insights from 28 years of Landsat data in southern Spain, Ecological Indicators, 171, 113 115, https://doi.org/10.1016/j.ecolind.2025.113115, 2025.

Hiebl, B., Alessi, N., Calvia, G., Bricca, A., Bonari, G., Zangari, G., Dorigo, W., Zerbe, S., and Rutzinger, M.: Advancing forest mapping: Pretraining strategies and deep-ensemble based uncertainty for predicting evergreen broad-leaved cover from Sentinel-2 time series, International Journal of Applied Earth Observation and Geoinformation, 142, 104 734, https://doi.org/10.1016/j.jag.2025.104734, 2025.

Hiebl, B., Alessi, N., Calvia, G., Bricca, A., Bonari, G., Zangari, G., Zerbe, S., and Rutzinger, M.: Combining specialized Sentinel-2 time series features with AlphaEarth Foundations for forest type mapping, ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences, XI-3-2026, 117–124, https://doi.org/10.5194/isprs-annals-XI-3-2026-117-2026, conference Name: XXV ISPRS Congress 2026 "From Imagery to Understanding", Commission III - 4–11 July 2026, Toronto, Canada, 2026.

Kang, Y., Ozdogan, M., Gao, F., Anderson, M. C., White, W. A., Yang, Y., Yang, Y., and Erickson, T. A.: A data-driven approach to estimate leaf area index for Landsat images over the contiguous US, Remote Sensing of Environment, 258, 112 383, https://doi.org/10.1016/j.rse.2021.112383, 2021.

Kapfer, J., Hédl, R., Jurasinski, G., Kopecký, M., Schei, F. H., and Grytnes, J.-A.: Resurveying historical vegetation data – opportunities and challenges, Applied Vegetation Science, 20, 164–171, https://doi.org/10.1111/avsc.12269, 2017.

Kennedy, R. E., Yang, Z., and Cohen, W. B.: Detecting trends in forest disturbance and recovery using yearly Landsat time series: 1. LandTrendr — Temporal segmentation algorithms, Remote Sensing of Environment, 114, 2897–2910, https://doi.org/10.1016/j.rse.2010.07.008, 2010.

Kollert, A., Bremer, M., Löw, M., and Rutzinger, M.: Exploring the potential of land surface phenology and seasonal cloud free composites of one year of Sentinel-2 imagery for tree species mapping in a mountainous region, International Journal of Applied Earth Observation and Geoinformation, 94, 102 208, https://doi.org/10.1016/j.jag.2020.102208, 2021.

Leonelli, G., Pelfini, M., Morra di Cella, U., and Garavaglia, V.: Climate Warming and the Recent Treeline Shift in the European Alps: the Role of Geomorphological Factors in High-Altitude Sites, AMBIO, 40, 264–273, https://doi.org/10.1007/s13280-010-0096-2, 2011.

Lobo, V. G. R., Fonseca, T. C. O., and Moura, F. A. S.: Bayesian cross-validation of geostatistical models, Spatial Statistics, 35, 100 394, https://doi.org/10.1016/j.spasta.2019.100394, 2020.

Mañas, O., Lacoste, A., Giró-i Nieto, X., Vazquez, D., and Rodríguez, P.: Seasonal Contrast: Unsupervised Pre-Training from Uncurated Remote Sensing Data, in: 2021 IEEE/CVF International Conference on Computer Vision (ICCV), pp. 9394–9403, ISSN 2380-7504, https://doi.org/10.1109/ICCV48922.2021.00928, iSSN: 2380-7504, 2021.

Mattioli, W., R, R., D, B., G, C., G, D., D, G., M, P., and P, C.: La Carta Forestale d'Italia (CFI2020): un ritratto aggiornato dei boschi italiani, Forest@ - Journal of Silviculture and Forest Ecology, 22, 39, https://doi.org/10.3832/efor4836-022, 2025.

Midolo, G., Clark, A. T., Chytrý, M., Essl, F., Dullinger, S., Jandt, U., Bruelheide, H., Dengler, J., Axmanová, I., Aćić, S., Argagnon, O., Biurrun, I., Bonari, G., Chiarucci, A., Ćušterevska, R., De Frenne, P., De Sanctis, M., Divíšek, J., Doležal, J., Dziuba, T., Ejrnæs, R., Garbolino, E., Jentsch, A., Jiménez-Alfaro, B., Lenoir, J., Moeslund, J. E., Napoleone, F., Rumpf, S. B., Svenning, J.-C., Swacha, G., Tatarenko, I., Večeřa, M., Vynokurov, D., and Keil, P.: Sixty years of plant community change in Europe indicate a shift toward nutrient-richer and denser vegetation, Science Advances, 12, eaeb2493, https://doi.org/10.1126/sciadv.aeb2493, 2026.

Miettinen, J., Breidenbach, J., Adame, P., Adolt, R., Alberdi, I., Antropov, O., Arnarsson, Ó., Astrup, R., Berger, A., Bogason, J., Chirici, G., Corona, P., D'Amico, G., Fejfar, J., Fischer, C., Gohon, F., Gschwantner, T., Hertzler, J., Koma, Z., Korhonen, K. T., Krajnc, L., Latte, N., Lejeune, P., McCullagh, A., Mionskowski, M., Moreno-Fernández, D., Myllymäki, M., Nilsson, M., Perin, J., Pitkänen, J., Redmond, J., Riedel, T., Schumacher, J., Seitsonen, L., Sirro, L., Skudnik, M., Snorrason, A., Sroga, R., Traub, B., Traustason, B., Westerlund, B., and Wurpillot, S.: Pan-European forest maps produced with a combination of earth observation data and national forest inventory plots, Data in Brief, 60, 111 613, https://doi.org/10.1016/j.dib.2025.111613, 2025.

Obuchowicz, C., Poussin, C., and Giuliani, G.: Change in observed long-term greening across Switzerland – evidence from a three decades NDVI time-series and its relationship with climate and land cover factors, Big Earth Data, 8, 1–32, https://doi.org/10.1080/20964471.2023.2268322, _eprint: https://doi.org/10.1080/20964471.2023.2268322, 2024.

Olofsson, P., Foody, G. M., Herold, M., Stehman, S. V., Woodcock, C. E., and Wulder, M. A.: Good practices for estimating area and assessing accuracy of land change, Remote Sensing of Environment, 148, 42–57, https://doi.org/10.1016/j.rse.2014.02.015, 2014.

Persson, H. J., Ekström, M., and Ståhl, G.: Quantify and account for field reference errors in forest remote sensing studies, Remote Sensing of Environment, 283, 113 302, https://doi.org/10.1016/j.rse.2022.113302, 2022.

Pflugmacher, D., Rabe, A., Peters, M., and Hostert, P.: Mapping pan-European land cover using Landsat spectral-temporal metrics and the European LUCAS survey, Remote Sensing of Environment, 221, 583–595, https://doi.org/10.1016/j.rse.2018.12.001, 2019.

Roy, D. P., Kovalskyy, V., Zhang, H. K., Vermote, E. F., Yan, L., Kumar, S. S., and Egorov, A.: Characterization of Landsat-7 to Landsat-8 reflective wavelength and normalized difference vegetation index continuity, Remote Sensing of Environment, 185, 57–70, https://doi.org/10.1016/j.rse.2015.12.024, 2016.

Rußwurm, M. and Körner, M.: Self-attention for raw optical Satellite Time Series Classification, ISPRS Journal of Photogrammetry and Remote Sensing, 169, 421–435, https://doi.org/10.1016/j.isprsjprs.2020.06.006, 2020.

Sabatini, F. M., Lenoir, J., Hattab, T., Arnst, E. A., Chytrý, M., Dengler, J., De Ruffray, P., Hennekens, S. M., Jandt, U., Jansen, F., et al.: sPlotOpen – An environmentally balanced, open-access, global dataset of vegetation plots, Global Ecology and Biogeography, 30, 1740–1764, https://doi.org/10.1111/geb.13346, 2021.

Safonova, A., Ghazaryan, G., Stiller, S., Main-Knorn, M., Nendel, C., and Ryo, M.: Ten deep learning techniques to address small data problems with remote sensing, International Journal of Applied Earth Observation and Geoinformation, 125, 103 569, https://doi.org/10.1016/j.jag.2023.103569, 2023.

Shannon, E. S., Finley, A. O., Hayes, D. J., Noralez, S. N., Weiskittel, A. R., Cook, B. D., and Babcock, C.: Quantifying and correcting geolocation error in spaceborne LiDAR forest canopy observations using high spatial accuracy data: A Bayesian model approach, Environmetrics, 35, e2840, https://doi.org/10.1002/env.2840, _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/env.2840, 2024.

Stewart, A. J., Robinson, C., Corley, I. A., Ortiz, A., Lavista Ferres, J. M., and Banerjee, A.: TorchGeo: Deep Learning With Geospatial Data, ACM Transactions on Spatial Algorithms and Systems, 11, 1–28, https://doi.org/10.1145/3707459, 2025.

Strunk, J. L., Reutebuch, S. E., McGaughey, R. J., and Andersen, H.-E.: An examination of GNSS positioning under dense conifer forest canopy in the Pacific Northwest, USA, Remote Sensing Applications: Society and Environment, 37, 101 428, https://doi.org/10.1016/j.rsase.2024.101428, 2025.

Sylvain, J.-D., Drolet, G., Thiffault, É., and Anctil, F.: High-resolution mapping of tree species and associated uncertainty by combining aerial remote sensing data and convolutional neural networks ensemble, International Journal of Applied Earth Observation and Geoinformation, 131, 103 960, https://doi.org/10.1016/j.jag.2024.103960, 2024.

Tan, J., Li, J., Ma, T., Yan, X., and Huo, Z.: Leveraging Sentinel-1/2 time series and deep learning for accurate forest tree species mapping, Frontiers in Forests and Global Change, 8, https://doi.org/10.3389/ffgc.2025.1599510, 2025.

Tong, X., Brandt, M., Yue, Y., Zhang, X., Fensholt, R., Ciais, P., Wang, K., Liu, S., Zhang, W., Mao, C., and Jepsen, M. R.: Reforestation policies around 2000 in southern China led to forest densification and expansion in the 2010s, Communications Earth & Environment, 4, 260, https://doi.org/10.1038/s43247-023-00923-1, 2023.

Tseng, G., Cartuyvels, R., Zvonkov, I., Purohit, M., Rolnick, D., and Kerner, H.: Lightweight, Pre-trained Transformers for Remote Sensing Timeseries, https://doi.org/10.48550/arXiv.2304.14065, arXiv:2304.14065, 2024.

Turubanova, S., Potapov, P., Hansen, M. C., Li, X., Tyukavina, A., Pickens, A. H., Hernandez-Serna, A., Arranz, A. P., Guerra-Hernandez, J., Senf, C., Häme, T., Valbuena, R., Eklundh, L., Brovkina, O., Navrátilová, B., Novotný, J., Harris, N., and Stolle, F.: Tree canopy extent and height change in Europe, 2001–2021, quantified using Landsat data archive, Remote Sensing of Environment, 298, 113 797, https://doi.org/10.1016/j.rse.2023.113797, 2023.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., and Polosukhin, I.: Attention Is All You Need, https://doi.org/10.48550/arXiv.1706.03762, arXiv:1706.03762 [cs], 2023.

Verheyen, K., De Frenne, P., Baeten, L., Waller, D. M., Hédl, R., Perring, M. P., Blondeel, H., Brunet, J., Chudomelová, M., Decocq, G., et al.: Combining Biodiversity Resurveys across Regions to Advance Global Change Research, BioScience, 67, 73–83, https://doi.org/10.1093/biosci/biw150, 2017.

Viana-Soto, A. and Senf, C.: The European Forest Disturbance Atlas: a forest disturbance monitoring system using the Landsat archive, Earth System Science Data, 17, 2373–2404, https://doi.org/10.5194/essd-17-2373-2025, 2025.

Wang, M., Zheng, Y., Huang, C., Meng, R., Pang, Y., Jia, W., Zhou, J., Huang, Z., Fang, L., and Zhao, F.: Assessing Landsat-8 and Sentinel-2 spectral-temporal features for mapping tree species of northern plantation forests in Heilongjiang Province, China, Forest Ecosystems, 9, 100 032, https://doi.org/10.1016/j.fecs.2022.100032, 2022.

White, J. C., Wulder, M. A., Hobart, G. W., Luther, J. E., Hermosilla, T., Griffiths, P., Coops, N. C., Hall, R. J., Hostert, P., Dyk, A., and Guindon, L.: Pixel-Based Image Compositing for Large-Area Dense Time Series Applications and Science, Canadian Journal of Remote Sensing, 40, 192–212, https://doi.org/10.1080/07038992.2014.945827, 2014.

Wulder, M. A., Roy, D. P., Radeloff, V. C., Loveland, T. R., Anderson, M. C., Johnson, D. M., Healey, S., Zhu, Z., Scambos, T. A., Pahlevan, N., Hansen, M., Gorelick, N., Crawford, C. J., Masek, J. G., Hermosilla, T., White, J. C., Belward, A. S., Schaaf, C., Woodcock, C. E., Huntington, J. L., Lymburner, L., Hostert, P., Gao, F., Lyapustin, A., Pekel, J.-F., Strobl, P., and Cook, B. D.: Fifty years of Landsat science and impacts, Remote Sensing of Environment, 280, 113 195, https://doi.org/10.1016/j.rse.2022.113195, 2022.

Yang, X., Qiu, S., Zhu, Z., Rittenhouse, C., Riordan, D., and Cullerton, M.: Mapping understory plant communities in deciduous forests from Sentinel-2 time series, Remote Sensing of Environment, 293, 113 601, https://doi.org/10.1016/j.rse.2023.113601, 2023.

Yel, S. G., Özmen, H. B., Öney Birol, S., Tunç Görmüş, E., and Kaplan, G.: Remote sensing applications for assessing climate change impacts on deciduous forests—A systematic review, Physics and Chemistry of the Earth, Parts A/B/C, 143, 104 321, https://doi.org/10.1016/j.pce.2026.104321, 2026.

Yuan, Y., Lin, L., Liu, Q., Hang, R., and Zhou, Z.-G.: SITS-Former: A pre-trained spatio-spectral-temporal representation model for Sentinel-2 time series classification, International Journal of Applied Earth Observation and Geoinformation, 106, 102 651, https://doi.org/10.1016/j.jag.2021.102651, 2022.

Yuan, Y., Lin, L., Xin, Q., Zhou, Z.-G., and Liu, Q.: An Empirical Study on Data Augmentation for Pixelwise Satellite Image Time-Series Classification and Cross-Year Adaptation, IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 18, 5172–5188, https://doi.org/10.1109/JSTARS.2025.3527017, 2025.

Yuan Yuan, Yuan, Y., Lin, L., and Lei Lin: Self-Supervised Pretraining of Transformers for Satellite Image Time Series Classification, 14, 474–487, https://doi.org/10.36227/techrxiv.13025039.v2, 2020.

Zanaga, D., Van De Kerchove, R., Daems, D., De Keersmaecker, W., Brockmann, C., Kirches, G., Wevers, J., Cartus, O., Santoro, M., Fritz, S., Lesiv, M., Herold, M., Tsendbazar, N.-E., Xu, P., Ramoino, F., and Arino, O.: ESA WorldCover 10 m 2021 v200, https://doi.org/10.5281/zenodo.7254221, 2022.

Zerveas, G., Jayaraman, S., Patel, D., Bhamidipaty, A., and Eickhoff, C.: A Transformer-based Framework for Multivariate Time Series Representation Learning, https://doi.org/10.48550/arXiv.2010.02803, 2020.

Zhu, Z. and Woodcock, C. E.: Continuous change detection and classification of land cover using all available Landsat data, Remote Sensing of Environment, 144, 152–171, https://doi.org/10.1016/j.rse.2014.01.011, 2014.